BBO capstone project presentation template

Develop a concise presentation that captures:

1. **An overview of your BBO approach**
   - Describe what you are trying to achieve or discover through the BBO capstone project. Focus on the core objective of your optimisation process, using clear and accessible language.
   - Briefly describe your overall strategy and how your process fits together.

WHAT I AM TRYING TO ACHIEVE
Eight unknown functions (2 to 8 inputs) are hidden behind a portal. I cannot see their formulas; I can only submit one input point per function per round and receive one number back. The goal is to find the input that gives the highest output for each function, using about 13 queries per function in total. Every query is expensive, so the real objective is to learn as much as possible from very few evaluations.

OVERALL STRATEGY
A repeatable weekly cycle:
1. Rebuild. The full history (175 supplied points plus all portal results) is rebuilt from the raw portal files every round, so nothing depends on memory or copy-paste.
2. Model. Fit a Gaussian-process (GP) surrogate per function - a statistical model that predicts the output and its uncertainty anywhere. Its settings (kernel type, smoothness) are re-selected each round by 5-fold cross-validation.
3. Classify. Put each function in one class: solved, noisy, still improving, or converged.
4. Choose a move to match the class: resubmit a known optimum; repeat a point to measure noise; take a GP-guided step; or move one coordinate to a fitted peak.
5. Predict and check. Write down a numerical prediction before submitting; the next result confirms or falsifies it.

All queries come from seeded Python scripts with a format validator, and the full history, datasheet and model card are kept in the GitHub repository (jhnlandman-sudo/CapStone).

2. **How your strategy has evolved**
   - Describe the key changes in your approach since your early rounds.
   - What influenced these changes – data trends, feedback, model performance or intuition?
   - Identify which principles or heuristics now guide your query decisions.

ROUNDS 1-3: GLOBAL SEARCH
One GP per function with an upper-confidence-bound (UCB) rule, plus simple moves away from the best supplied points. Useful for finding promising regions, but blind to how each function was shaped.

ROUNDS 4-7: LOCAL STRUCTURE
Once good regions emerged I switched to cheaper, more exact tools: moving one coordinate at a time and fitting a three-point quadratic to find its peak; a closed-form Gaussian fit that solved F1 exactly; trust-region steps on F6/F7. From round 7 the GP settings were tuned by cross-validation instead of fixed.

ROUNDS 8-12: NOISE AND DISCIPLINE
Three confident predictions on F2/F3 failed. Repeating the same input showed why: F2 returned 0.650, 0.726 and 0.598 at one point - it is noisy, and I had been fitting curves to noise. Since then repeats are a deliberate tool, and each function is classified before choosing a move.

WHAT DROVE THE CHANGES
Mainly model performance against my own written predictions (falsified predictions forced the noise discovery), then data trends (which coordinates still paid).

HEURISTICS I NOW FOLLOW
- Write the prediction before the query.
- Never fit a curve to differences smaller than the noise.
- Replicate before trusting a single high reading.
- In high dimensions move one coordinate at a time; joint GP steps failed on F7 and F8.
- Use the GP to rank candidates, not to predict levels.
- Spend queries by expected gain, not by function.

3. **Patterns, data and insights**
   - Highlight the most meaningful trends you've observed across your data points or model evaluations.
   - Identify which variables or behaviours seem to influence results the most.
   - Explain how these observations have shaped your understanding of the search process.

BEST RESULTS (after round 11: best supplied point -> my best)
- F1: 0 -> 2.000 (solved: Gaussian peak)
- F2: 0.611 -> 0.726 (noisy ridge)
- F3: -0.035 -> -0.025 (flat and noisy)
- F4: -4.03 -> 0.744 (very sharp peak)
- F5: 1,089 -> 8,662 (solved: corner 1,1,1,1)
- F6: -0.714 -> -0.061 (narrow peak)
- F7: 1.365 -> 2.301 (still improving)
- F8: 9.598 -> 9.925 (converged)

MOST MEANINGFUL TRENDS
- Three kinds of function: noise-free (F1, F5 repeat exactly; F7, F8 fit with near-zero noise), noisy (F2 noise ~0.06, F3 and F6 ~0.01-0.02) and very sharp (F4 curvature about -2,700 along x4).
- Diminishing returns: F7's single-coordinate steps gained +0.106, +0.075, +0.046, then flattened.
- Threshold behaviour: F5 rose slowly with each input, then jumped 84% when all four inputs reached their upper bound together.

WHICH VARIABLES MATTER
The GP's per-input lengthscales act like a variance breakdown:
- F2: x1 explains 98%; x2 barely matters (a ridge near x1 = 0.68).
- F3: almost everything is x3.
- F7: x5, x4 and x2 dominate; x1 and x3 together about 6%.
- F8: three of eight inputs (x3, x1, x7) carry about 70%.

A PCA of my own F8 queries showed most of my sampling went along the two least relevant inputs.

WHAT THIS TAUGHT ME
The search space is mostly low-dimensional in practice. Fewer, better-chosen directions beat more exploration, and a single good reading is a hypothesis, not a result.

4. **Decision-making and iteration**
   - Explain how you balance exploration and exploitation.
   - Describe one or two examples of strategic decisions. What worked, what didn't, and why?
   - Reflect briefly on how you handle uncertainty or adapt when results don't match expectations.

BALANCING EXPLORATION AND EXPLOITATION
Exploration is worth most early and is worth nothing in the final round, so my balance shifted deliberately over time: UCB with a moderate exploration weight (beta 0.5-0.8) early; trust regions that shrink as functions converge; and at the end pure exploitation. Because the competition keeps the best-so-far value, a probe that scores badly costs information, not points - which makes cheap probes worthwhile until late.

EXAMPLE 1 - WHAT WORKED: F5 CORNER
Each input improved steadily as it was pushed upward. Rather than continue small steps, I tested the hypothesis that the optimum lies at the corner (1,1,1,1). Result 8,662 against 4,718 (+84%). A second probe (x2 = 0.9) fell to 7,073, confirming the corner, and a repeat returned 8,662.405 exactly - so F5 is solved and noise-free.

EXAMPLE 2 - WHAT DIDN'T: F7 SIX-COORDINATE STEP
A GP-guided step that moved all six inputs together gained +0.133 in round 10. Repeating the same kind of move in round 11 was predicted to give 2.447 but returned 2.202 - nearly five standard deviations below. The GP is least reliable in directions where it has little data, and a six-dimensional step goes exactly there. Adjustment: back to single-coordinate moves on F7 and F8.

HANDLING UNCERTAINTY
Every surprise is classified as noise, model error or a real feature, using repeats and the written predictions. Noisy functions are judged on averages; a falsified prediction changes the method, not just the next point.

5. **Next steps and reflection**
   - Outline your planned actions or experiments to improve the model's performance.
   - Include two or three sentences connecting your work to the broader ML landscape.
   - Add a short statement on how you'd communicate your results to a non-technical audience or stakeholder.

PLANNED ACTIONS
- Round 13 (final) is pure exploitation: F1 and F5 resubmit their known optima; F4, F7, F8 keep this round's single-coordinate probe only if it improved; F2, F3, F6 go to the point with the best average reading, not the luckiest single one.
- Record every prediction in a machine-readable log so prediction accuracy can be calculated, not just described.
- Upload the full history, scripts, datasheet and model card to GitHub so every query can be reproduced.

BROADER ML CONNECTION
This is Bayesian optimisation, the same approach used for hyperparameter tuning, engineering design and drug discovery whenever each evaluation is expensive. The capstone also illustrated wider lessons: models need explicit noise handling, high-dimensional problems are effectively low-dimensional, and a method is only trustworthy if its predictions are tested against outcomes. My direct application is calibrating expensive actuarial projection models, where each run takes minutes to hours.

COMMUNICATING TO A NON-TECHNICAL AUDIENCE
I would describe it as tuning eight black-box machines with only thirteen tries each: first learn which dials matter, then home in carefully, and test before trusting a lucky result. I would show one chart of best result by round for each function, and summarise each as solved, improving or limited by noise - with a plain statement of what we know for sure and what is still uncertain.